

Anup Barve

✉ anup5barve@gmail.com ☎ 612-449-9199 🏠 [Google Scholar](#) [in anupvbarve](#)

Professional Summary

PhD candidate in Mechanical Engineering (atmospheric science focus) at the University of Minnesota, specializing in atmospheric boundary layer turbulence and high-fidelity computational fluid dynamics. My research develops a mesoscale-to-microscale coupled framework linking regional-scale weather to boundary-layer-resolving large-eddy simulation with Lagrangian/bulk cloud microphysics, validated against in-situ and remote-sensing observations from two international field campaigns (FATIMA, MAGPIE). Experienced in HPC-based solver development (FORTRAN/MPI), extreme weather simulation, and leading peer-reviewed publications from conception through publication.

Education

University of Minnesota

Ph.D. Mechanical Engineering

Minneapolis, USA

Sep 2021 – July 2026

- **Dissertation:** Exploring cloud and fog processes through multi-scale coupled modeling frameworks.
- **Focus:** High-fidelity CFD (large-eddy simulation), convective and stratiform cloud systems, turbulence, precipitation processes, mesoscale-to-microscale coupling, and aerosol–cloud interactions.
- **Advisor:** Prof. Lian Shen

Indian Institute of Technology, Madras

M.Tech. Mechanical Engineering

Chennai, India

July 2017 – May 2019

- **Thesis:** Numerical study of particle-laden coaxial turbulent jets.
- **Focus:** Turbulence modeling, multiphase flows, Eulerian–Lagrangian methods, subgrid-scale parameterizations.
- **Advisors:** Prof. Kameswararao Anupindi & Prof. Srikrishna Sahu

Goa University

B.E. Mechanical Engineering

Goa, India

July 2010 – May 2014

Research Experience

University of Minnesota

Graduate Research Assistant

Minneapolis, USA

Sep 2021 – Present

- Developed a high-performance parallel FORTRAN/MPI large-eddy simulation (LES) solver for simulating atmospheric boundary layer flows, incorporating SLURM-based workflow automation for scalable HPC deployment.
- Maintained version control of the LES solver and analysis codebase using Git/GitHub throughout development.
- Implemented Lagrangian and bulk cloud microphysics parameterizations to simulate cloud droplet activation, condensation, coalescence, precipitation formation, and cloud/fog life cycle evolution.
- Designed and deployed a mesoscale-to-microscale coupled framework spanning regional-scale weather (large-scale dynamics), boundary-layer-resolving LES, and Lagrangian cloud microphysics to simulate realistic marine atmospheric boundary layer phenomena.
- Built Python-based dynamic downscaling workflows using ERA5 reanalysis and COAMPS numerical weather prediction data to generate high-resolution forcing for cloud-resolving and convection-permitting simulations.
- Demonstrated the capability of the coupled Large-Scale Dynamics–LES framework to resolve deep convection over Barbados associated with the African Easterly Wave that operational global models failed to capture.
- Validated model simulations against multi-platform observational datasets, including surface energy fluxes, precipitation records, radar reflectivity, and lidar retrievals, to evaluate physical and microphysical process representation.
- Implemented bulk ice microphysics parameterizations to simulate ice fog and mixed-phase cloud processes

in high-latitude environments, and validated simulation results against observational data from the Atmospheric Radiation Measurement (ARM) North Slope of Alaska (NSA) site.

- Participated in the FATIMA and MAGPIE international field campaigns, collaborating with peers on proposal development, simulation planning, data analysis, sponsor review meetings, and peer-reviewed publications.
- Contributed to proposal development for externally funded research projects with collaborators from the Earth system science community.

IIT Madras

Graduate Research Assistant

Chennai, India

May 2018 – May 2019

- Conducted LES and RANS simulations of particle-laden coaxial turbulent jets using OpenFOAM, investigating turbulence–particle interactions.
- Developed and modified subgrid-scale turbulence parameterizations (subgrid kinetic energy model) to improve representation of turbulent mixing in multiphase flow simulations.
- Applied an Eulerian–Lagrangian framework to study two-way coupling between the carrier phase and dispersed particles.
- Validated numerical simulations against experimental benchmarks and performed parametric studies of particle dispersion and turbulent transport under varying flow conditions.
- This work resulted in a peer-reviewed publication selected as an Editor’s Pick.

Technical Skills

Programming Languages: Python, FORTRAN, C/C++, MATLAB

High-Performance Computing: MPI, OpenMP, SLURM, bash scripting

AI/ML Methods: Deep learning (PyTorch), surrogate modeling, Bayesian optimization, neural network-based prediction frameworks, LLM-assisted software development

Atmospheric & Earth System Modeling: Large-eddy simulation (LES), cloud microphysics parameterization (Lagrangian & bulk), ERA5 reanalysis, COAMPS

CFD & Simulation Software: OpenFOAM, ANSYS Fluent, custom solver development

Numerical Methods: Finite difference/volume methods, reduced-order modeling, uncertainty quantification

Data Analysis & Visualization: Observational data analysis (radar, lidar, surface flux), statistical methods, ParaView, Tecplot, Gnuplot

Tools & Version Control: CMake, Make, Git/Github, Linux, VSCode

Selected AI/ML Projects

AI/ML Surrogate Modeling for Physics-Based Simulation with Uncertainty Quantification

- Developed data-driven surrogate models (reduced-order models) to rapidly predict aerodynamic quantities across a continuous parameter space, replacing expensive physics-based simulations.
- Automated simulation workflows using cloud-based APIs and incorporated uncertainty quantification to estimate prediction confidence across the design range.

Skills: Surrogate/Reduced-Order Modeling, Uncertainty Quantification, Python, Workflow Automation

AI-Based Joint Prediction Modeling on QMNIST

- Built a balanced and quality-controlled QMNIST dataset by grouping images by writer and digit class, performing preprocessing, validation, and structured batching for AI training.
- Designed a multi-path neural architecture with shared embeddings and masking logic to enforce predictive consistency, documenting the workflow and loss decomposition.
- Trained and evaluated the model using cross-entropy and constraint-based objective functions, analyzing performance trends, and verifying consistency across prediction paths.

Skills: PyTorch, Multi-Task Learning, Neural Architecture Design

Publications

Lee, S. et al. (2026). “A review of the FATIMA Yellow Sea field campaign research”. In: *Quarterly Journal of the Royal Meteorological Society*. e70157 QJ-24-0380, e70157. DOI: [10.1002/qj.70157](https://doi.org/10.1002/qj.70157) .

Barve, A. et al. (2025b). “Large-eddy simulation with Lagrangian cloud modeling and large-scale dynamics (L^3) for studying the marine fog life cycle”. In: *Quarterly Journal of the Royal Meteorological Society* 151.772, e5022. DOI: [10.1002/qj.5022](https://doi.org/10.1002/qj.5022) .

Fernando, H. J. et al. (2025). “FATIMA-GB: Searching Clarity within Marine Fog”. In: *Bulletin of the American Meteorological Society* 106.6, E971–E1016. DOI: [10.1175/BAMS-D-23-0050.1](https://doi.org/10.1175/BAMS-D-23-0050.1) .

Barve, A. V. et al. (2020). “Effect of co-flow velocity ratio on evolution of poly-disperse particles in coaxial turbulent jets: A large-eddy simulation study”. In: *Physics of Fluids* 32.9. DOI: [10.1063/5.0017663](https://doi.org/10.1063/5.0017663) .

Articles in Preparation

Barve, A. et al. (2026a). *Investigating stratus cloud lowering over the Yellow Sea using large-eddy simulation*. In preparation for Weather and Forecasting.

Barve, A. et al. (2026c). *Investigating the impact of African Easterly Waves on the marine atmospheric boundary layer*. In preparation for Boundary-Layer Meteorology.

Barve, A. et al. (2026e). *Large-eddy simulation of ice fog lifecycle over the North Slope of Alaska*. In preparation for Quarterly Journal of the Royal Meteorological Society.

Mittal, S. et al. (2026). *Impact of coastal topography on marine boundary-layer structure and cloud formation at Ragged Point, Barbados*. In preparation for Boundary-Layer Meteorology.

Conference Publications

Barve, A. et al. (2026b). “Investigating the effects of the African Easterly Wave on the marine atmospheric boundary layer using large-eddy simulation”. In: *106th AMS Annual Meeting*. AMS.

Barve, A. et al. (2026d). “Investigating the sensitivity of stratus lowering marine fog using large-eddy simulation”. In: *106th AMS Annual Meeting*. AMS.

Gultepe, I. et al. (2026). “A short review on fog and integrated visibility”. In: *106th AMS Annual Meeting*. AMS.

Barve, A. et al. (2025a). “Bridging scales in marine fog: coupling large-eddy simulation with large-scale dynamics and microphysics”. In: *AGU25*.

Gultepe, I. et al. (2025a). “Marine fog microphysics and dynamics during the FATIMA GB and YS campaigns”. In: *AGU25*.

Gultepe, I. et al. (2025b). “Marine fog research during FATIMA field campaigns”. In: *105th Annual AMS Meeting 2025*.

Barve, A. et al. (2024). “Investigation of stratus-fog transitions over Yellow Sea using large-eddy simulation coupled with large-scale dynamics and Lagrangian cloud model”. In: *AGU24*.

Gultepe, I. et al. (2024). “Fog microphysics and its parameterizations during FATIMA marine fog project”. In: *AGU Fall Meeting Abstracts*. Vol. 2024, A53V–06.

Mittal, S. et al. (2024). “Investigating the impact of coastal topography on marine fog dynamics using large-eddy simulations”. In: *APS Division of Fluid Dynamics Meeting Abstracts*, ZC26–007.

Barve, A. et al. (2023). “Effect of radiative fluxes on the life cycle of marine fog: a large-eddy simulation study”. In: *AGU23*.

Awards

Best AOFD Student Poster Presentation <i>American Meteorological Society (106th AMS Annual Meeting)</i>	2026
Adrian ME Graduate Student Travel Award <i>Department of Mechanical Engineering, University of Minnesota</i>	2026
Charles C. S. Song Fellowship <i>St. Anthony Falls Laboratory, University of Minnesota</i>	2025
Ministry of Human Resources & Development Fellowship <i>Indian Institute of Technology, Madras</i>	2017-2019
Graduate Aptitude Test in Engineering <i>Achieved a 99.73 percentile rank among 200,000 students</i>	2016

Teaching Experience

University of Minnesota <i>Teaching Assistant</i>	<i>Minneapolis, USA</i>
◦ Led recitation sessions for Thermodynamics (Spring 2026), supporting student learning in core concepts, problem solving, and exam preparation. ◦ Led discussion sessions for Fluid Mechanics (Fall 2022), helping students strengthen conceptual understanding and analytical problem-solving skills. ◦ Supervised laboratory sessions for Mechanical Measurements Laboratory (Spring 2022), guiding students in experimental methods, data analysis, and technical reporting.	
IIT Madras <i>Teaching Assistant</i>	<i>Chennai, India</i>
◦ Assisted instruction in Thermal Engineering Laboratory (2019), supporting student learning in thermodynamics, heat transfer, and laboratory practice.	

Professional Experience

Siemens Ltd. <i>Senior Executive</i>	<i>Goa, India</i> <i>March 2020 – June 2021</i>
◦ Led the development and testing of an electrical switchgear prototype to replace high-global-warming-potential SF₆-based systems, collaborating with interdisciplinary global teams on sustainable engineering solutions.	
Baker Hughes <i>ASPIRE Development Program</i>	<i>Bangalore, India</i> <i>July 2019 – Feb 2020</i>
◦ Contributed to Baker Hughes' net-zero mission by developing machine learning predictive models for early gas-turbine failure detection, reducing downtime, replacement needs, and associated cost, transport, and CO₂ emissions through collaboration with interdisciplinary global teams.	
NLC India Ltd. <i>Graduate Engineer Trainee</i>	<i>Tuticorin, India</i> <i>Dec 2016 – June 2017</i>
◦ Operated a thermal power plant with responsibility for safety, reliability, and optimal grid-load management in a large-scale energy system.	
A. W. Faber-Castell Pvt. Ltd. <i>Engineer Trainee</i>	<i>Goa, India</i> <i>Dec 2014 – June 2015</i>
◦ Led maintenance operations and mentored innovation-driven process-improvement projects, building early experience in leadership, teamwork, and technical problem-solving.	